

Yuvraj Singh

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EDUCATION

Malaviya National Institute of Technology, Jaipur

Bachelor of Technology

Jaipur, Rajasthan

Nov 2020 – May 2024

EXPERIENCE

Software Developer Engineer

June 2025 – Present

Amazon Development Centre

- Implemented pagination for a high-traffic e-commerce fulfillment API to support a logistics placement feature, ensuring **backward compatibility** via zero-downtime deployment across multi-region services. **Impact:** Expanded feature eligibility to ~15% of inbound volume, achieving **+250 bps conversion uplift** and contributing to **\$21M+ projected cost savings**.
- Owned enforcement workflow implementation in a seller-facing portal, integrating document storage, building **file validation** (PDF-only, 10MB limit), and configuring **CloudWatch alarms** with load testing up to **1000 TPS**. **Impact:** Protected **\$5B+** in seller merchandise across **59K+** sellers under compliance risk.
- Built core **CRUD APIs** for a centralized artifact repository, including API contract design, metadata retrieval, **presigned URL generation**, authentication onboarding for containerized services, cross-account DNS migration, and integration testing. **Impact:** Delivered foundational APIs supporting **850K+ documents** across **50+ compliance use cases** with a **99.9% availability target**.

Software Developer

Aug 2024 – May 2025

Ruppy - CarDekho Group

- Designed and developed backend APIs using **Node.js** and **Express.js**, integrating **Digilocker**, **CIBIL APIs**.
- Built user journey screens for loans and refinancing supporting **150K+ users**.
- Led migration from **AWS Lambda** to **ECS Fargate**, improving scalability by **40%** and reducing costs by **15%**.
- Optimized performance by reducing **MongoDB load** by **60%** using caching.

PROJECTS

Vyapaar House | [Live](#) | [GitHub](#)

- Architected **3 decoupled Spring Boot microservices** (Order, Stock, Email) using **RabbitMQ topic exchange**, enabling parallel asynchronous stock deduction and email confirmation on payment with zero inter-service coupling; containerized via **Docker**.
- Built a full-stack **Indian textile marketplace** using **React 18**, **Supabase**, and **PostgreSQL**, featuring **Google OAuth**, **Razorpay INR payments**, real-time order tracking, and **Hindi + English full-text search** using **tsvector GIN index**.
- Implemented **dual rate limiting**: **Token Bucket** on AI endpoints using **Vercel KV (Redis)** and **Sliding Window** on order services using **ConcurrentHashMap**, enforcing per-IP and per-user request limits.
- Integrated a **Claude AI shopping agent** via **Vercel Edge Functions**, enabling natural Hindi/English conversations for product discovery, cart management, and seamless checkout.

SKILLS

Languages: Java, JavaScript, C++

Frameworks & Libraries: Spring Boot, React.js, Node.js, Express.js

Databases: PostgreSQL, MongoDB, MySQL, Redis, DynamoDB, Supabase

Cloud (AWS): EC2, ECS, Lambda, S3, SQS, SNS, DynamoDB, CloudWatch, IAM, API Gateway

Messaging & Event-Driven: RabbitMQ, SQS, SNS, Event-Driven Architecture, Async Messaging

DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, Vercel, Render

Auth & Payments: JWT, OAuth 2.0, Supabase Auth, Razorpay

AI & Search: Claude API (Tool Use), Postgres Full-Text Search, Vercel KV (Redis)

Concepts: Microservices, Rate Limiting, System Design, RESTful APIs, OOPS

Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning

ACHIEVEMENTS

- Achieved **6 in DSA** and **5 in C++** on HackerRank.